

UPSC Prelims

Environment & Ecology

Class No. 2

Basics of Environment part 2

Examples of biotic interactions



- Mutualism
- Ants and aphids: Ants "farm" aphids by protecting them from predators and moving them to new food sources. In return, the aphids secrete a sweet substance called honeydew that the ants eat.
- Pollination: Bees, butterflies, and other insects feed on the nectar and pollen of flowers. In the process, they transfer pollen from one flower to another, aiding in plant reproduction.

Examples of biotic interactions



- Symbiosis
- Mycorrhizae: These are symbiotic associations between fungi and the roots of plants. The fungi help the plants absorb nutrients from the soil, while the plants provide the fungi with carbohydrates.
- Coral and zooxanthellae: Coral reefs are formed by the symbiotic relationship between corals and a type of algae called zooxanthellae. The algae provide the coral with food through photosynthesis, while the coral provides the algae with a protective environment and access to sunlight.

Examples of biotic interactions



- Proto-Cooperation :
- Lions and hyenas: Although lions and hyenas are usually considered to be enemies, they will sometimes cooperate to steal a kill from another predator. This allows both species to benefit from the shared food source.

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- Birds and grazing animals: Some species of birds will follow grazing animals, such as cattle or bison, to feed on insects that are disturbed by the animal's movement. This benefits both the birds, who get access to a food source, and the grazing animals, who get rid of insects that can harm their skin or eyes.

Examples of biotic interactions



- Parasitism:
- Ticks and mammals: Ticks are blood-sucking parasites that feed on the blood of mammals, such as deer and humans. They benefit from the host's blood, while the host can be harmed by the loss of blood and potential transmission of diseases.
- Dodder and plants: Dodder is a parasitic plant that attaches itself to other plants and steals nutrients and water from them. It benefits from the host's resources, while the host can be harmed by the loss of nutrients and potential damage to its tissues.

Examples of biotic interactions



- Commensalism:
- Barnacles and whales: Barnacles are marine crustaceans that attach themselves to the skin of whales. They benefit from the movement of the whale, which helps them obtain food, while the whale is not significantly affected by their presence.
- Epiphytes and trees: Epiphytes are plants that grow on other plants, such as trees. They benefit from the structure of the tree, which provides support and access to sunlight, while the tree is not significantly affected by their presence.

Examples of biotic interactions



- Amensalism:
- Black walnut trees and other plants: Black walnut trees release a chemical called juglone that inhibits the growth of other plants growing near them. This inhibitory effect is an example of amensalism.
- Antibiotic-producing bacteria and non-producing bacteria: Some bacteria produce antibiotics that inhibit the growth of other bacteria in the same environment. This inhibition can be considered an example of amensalism.

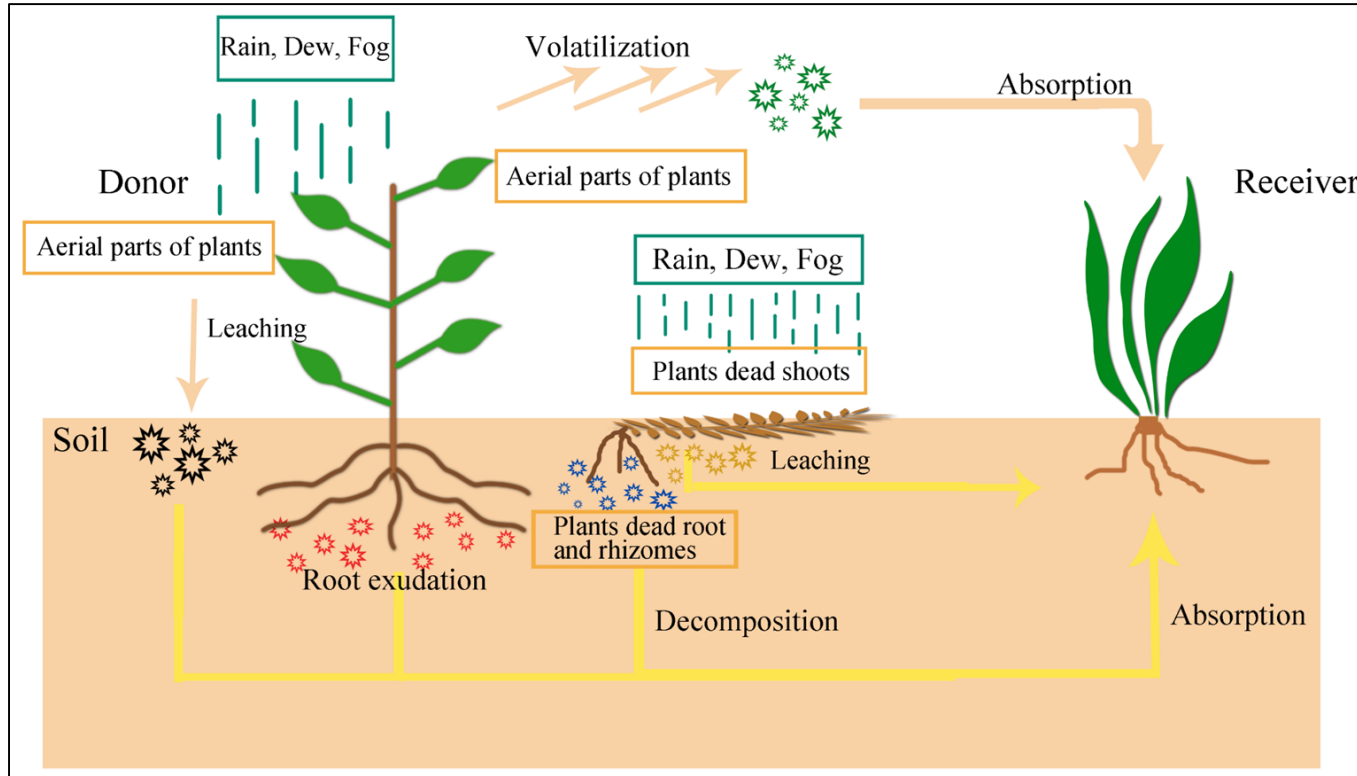
Examples of biotic interactions



- Competition:
- Lions and hyenas: Lions and hyenas both hunt and scavenge for food in the same areas. As they compete for the same resources, they can engage in fierce competition.
- Trees in a forest: Trees in a forest often compete for light, water, and nutrients. As they grow taller, they can shade out and deprive other trees of sunlight and nutrients.

- Allelopathy is a type of chemical interaction between plants in which one plant releases chemicals or secondary metabolites that affect the growth, survival, or reproduction of another plant.
- The chemicals produced by one plant species can either be harmful or beneficial to another plant species.
- These chemicals can be released through leaching, volatilization, or decomposition of plant residues.

ALLELOPATHY



- Allelopathy can have both negative and positive effects on plant growth and survival.
- Some plants produce chemicals that inhibit the growth of other plants, which can help them to compete for resources such as water, nutrients, and light.
- Other plants produce chemicals that have a positive effect on the growth of other plants, such as the production of growth hormones.
- Example: Black walnut trees release juglone, a chemical that is toxic to many plants and can inhibit their growth. Similarly, eucalyptus trees produce chemicals that can inhibit the growth of many other plant species, allowing them to dominate the surrounding vegetation.

- Allelopathy
- Sunflowers: Sunflowers release chemicals called allelochemicals, which can inhibit the growth of other plants growing nearby. This chemical effect can help sunflowers outcompete other plants for resources.
- Eucalyptus trees: Eucalyptus trees release allelochemicals that inhibit the growth of many other plant species. This effect can help eucalyptus trees outcompete other plants in their environment

NEUTRALISM



- Neutralism is also sometimes described as the relationship between two species inhabiting the same space and using the same resources, but that have no effect on each other.
- There is no net benefit or harm to either species.
- Perhaps in some interspecific interactions, the costs and benefits experienced by each partner are exactly the same so that they sum to zero. It is not clear how often this happens in nature.

Ecosystem Productivity



- Biomass generation or stored energy in an ecosystem per unit area per unit time
 - Units- $\text{gm/m}^2/\text{year}$ or $\text{kcal/m}^2/\text{year}$
- Primary productivity
 - Rate of biomass production by autotrophs

- Divided into two
 - **Gross primary productivity** of an ecosystem is the rate of production of organic matter during photosynthesis.
 - **Net primary productivity** is Gross primary productivity minus respiration losses (R)
 - $NPP = GPP - R$
- Secondary productivity
 - Rate of formation of new organic matter by consumers.
- The annual net primary productivity of the whole biosphere is approximately 170 billion tons (dry weight) of organic matter
 - Oceans- 32% vs Land- 68%

Ecosystem Productivity : How to check?



- Lower latitudes > higher latitudes : more sunlight in tropics
- Lower altitudes > higher altitudes : more sunlight
 - Sunlight is an important factor
- Continents > ocean : per unit area, ocean has very small photic zone.
- Arctic > Antarctic
- NPP measured using dry weight (biomass)
 - Removes effect of season as moisture varies in the year

Ecosystem Productivity



- Important pointers
- Higher productivity regions
 - Tropical forests, ecotones regions- estuaries, wetlands, mangroves, coral reefs
- Medium productivity regions
 - Temperate forests, grasslands (tropical > temperate), savannah > steppes, freshwater lakes, farm lands, ocean bottoms.

Ecosystem Productivity



- Low productivity regions : desert, permafrost, mountain peaks, arctic > antarctic.
- Tropical Rainforests contribute 22% of total NPP and open ocean about 24%. Open ocean has a far lesser rate of NPP, but covers a far larger area.
- Overall, terrestrial ecosystems contribute 66% while marine ecosystems contribute 34%
- Arctic > antarctic : arctic is just ice floating over sea, not a landmass. Gets more sunlight, Arctic Amplification.

Ecosystem Productivity



ecosystem	Net primary production - dry ($\text{g m}^{-2}\text{year}^{-1}$)	Area (10^6km)
Tropical forest	2,000	20
Temperate forest	1,300	18
Boreal forest	800	12
Savannah	700	15
Temperate grassland	500	9
Tundra & alpine	140	8
Desert scrub	70	18
Extreme desert (sand/rock/ice)	3	24
Agricultural land	650	14
Open ocean	125	332
Continental shelf	350	27
Coral reef & estuaries	2,000	2
Overall for planet earth	320	510

Ecosystem Productivity



Ecosystem	Average NPP
Swamps and Marshes	2000
Streams and Lakes	500
Scrub Forest	600
Total Terrestrial Ecosystem	730
Total Marine ecosystem	155
Ecotone areas	Generally higher

Ecosystem Productivity

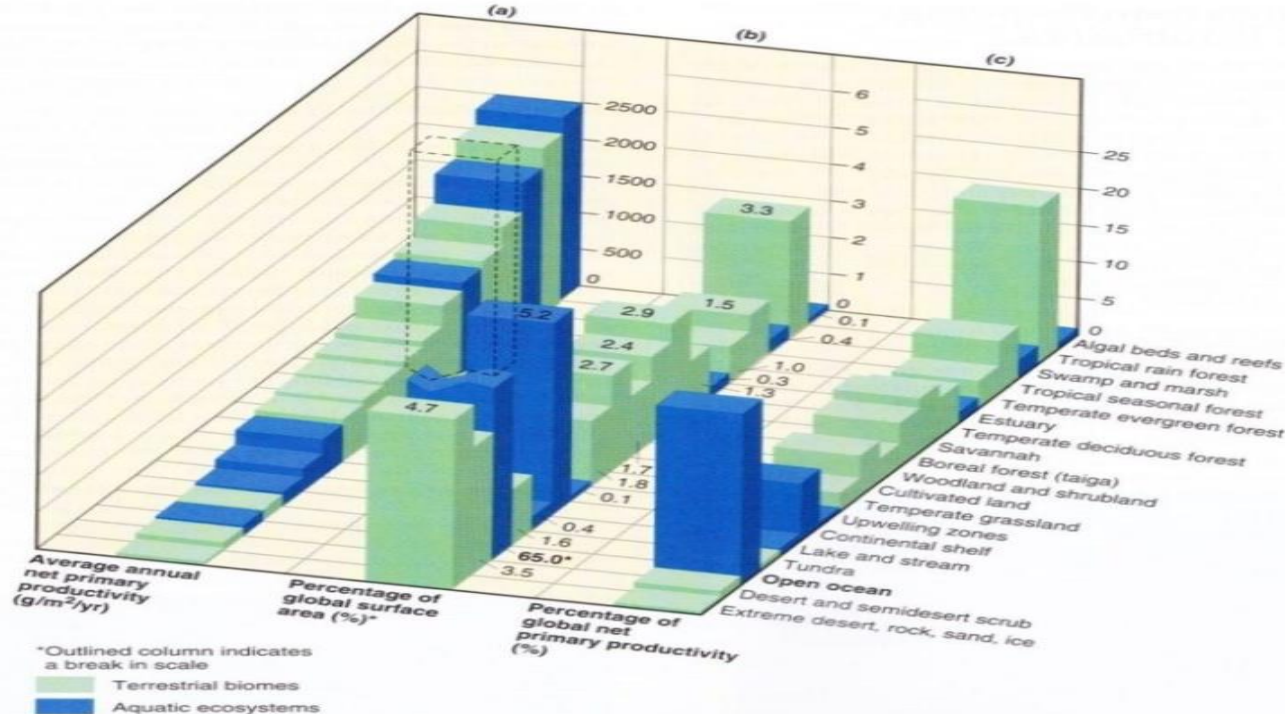


Figure 5-13 Productivity of different ecosystems. (a) The annual net primary productivity of different ecosystems, (b) the percentage of different ecosystems over Earth's surface area, and (c) the percentage of global net primary productivity. The outlined column for open ocean in (b) shows that the open ocean is such a high percentage of Earth's surface that it is off the scale relative to all of the other ecosystems.

ECOSYSTEM COLLAPSE



- It refers to phenomenon when an ecosystem rapidly lose their structure and function, with dramatic changes to their size or extent, or the species that comprise them.
- These losses tend to homogenise and simplify the ecosystem — fewer species, fewer habitats and fewer connections between the two.
- It can lead to catastrophic declines of carrying capacity and mass extinction, and can also pose existential risk to human populations.

ECOSYSTEM COLLAPSE



Australia Fire (2019-2020)



American Dust Bowl

Ecological Succession



- An important characteristic of all communities is that their composition and structure constantly change in response to the changing environmental conditions.
- Ecological succession- gradual and fairly predictable change in the species composition of a given area
- During succession some species colonise an area and their population become more numerous whereas populations of other species decline and even disappear.

Ecological Succession



- The changes are gradual and fairly predictable. This is called succession.
- These changes finally lead to a community that is in equilibrium with the environment called climax community.

Ecological Succession



- Sere- entire sequence of communities that successively change in a given area.
- Seral communities- individual transitional communities
- Sequence
- Primary species > seral communities > climax community
- Primary succession
- Occurs in areas where no living organisms ever existed, say bare rock, volcanic eruption, lakes etc
- Secondary succession
- Begins in areas where natural biotic communities have been destroyed such as in abandoned farm lands, burned or cut forests, lands that have been flooded.

Ecological Succession



- Primary succession : Newly cooled lava, bare rock, newly created pond, reservoir. Establishment of a new biotic community is generally slow. Before a biotic community is established, soil is needed.
- Depending on climatic conditions, soil formation can take many years.
- Secondary succession : in areas where natural biotic communities have been destroyed. Some soil is already present, therefore it is faster.
- As succession is proceeding, it is possible that human or natural induced disturbance could convert a particular seral stage into a previous stage.

Ecological Succession in plants



- Wet areas : known as hydrarch succession and the progress is from hydrarch to mesic conditions.
- Dry areas : xerarch succession and the progress is from xeric to mesic conditions.
- Therefore the final conditions are the same : mesic (medium water conditions)

Primary Succession



- Dry conditions :
- Species that invade a bare area : pioneer species.
- These species are generally lichens which are able to secrete acids to dissolve rocks helping in weathering and soil formation. They pave the way for mosses (bryophytes) that take hold with small amount of soil.
- Later, stable climax community is formed.
- Wet Conditions : Pioneers are phytoplankton, followed by rooted submerged plants, followed by rooted floating angiosperms, marsh and swamp, scrubs and finally trees.

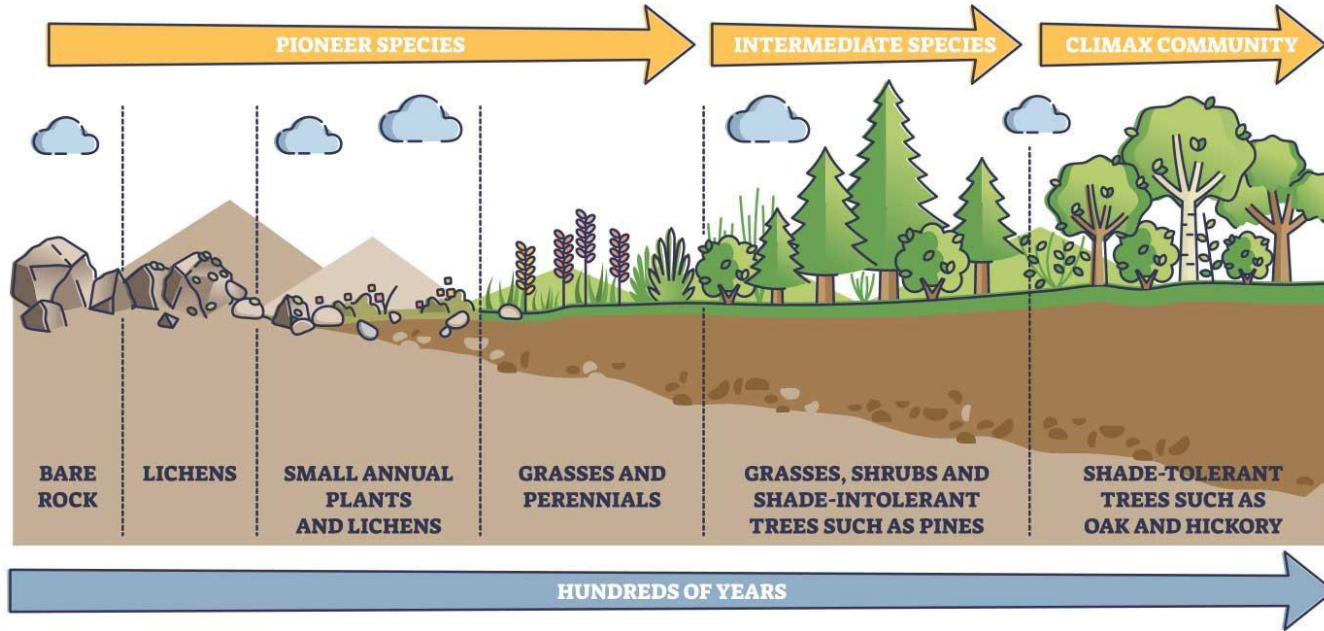
Secondary Succession



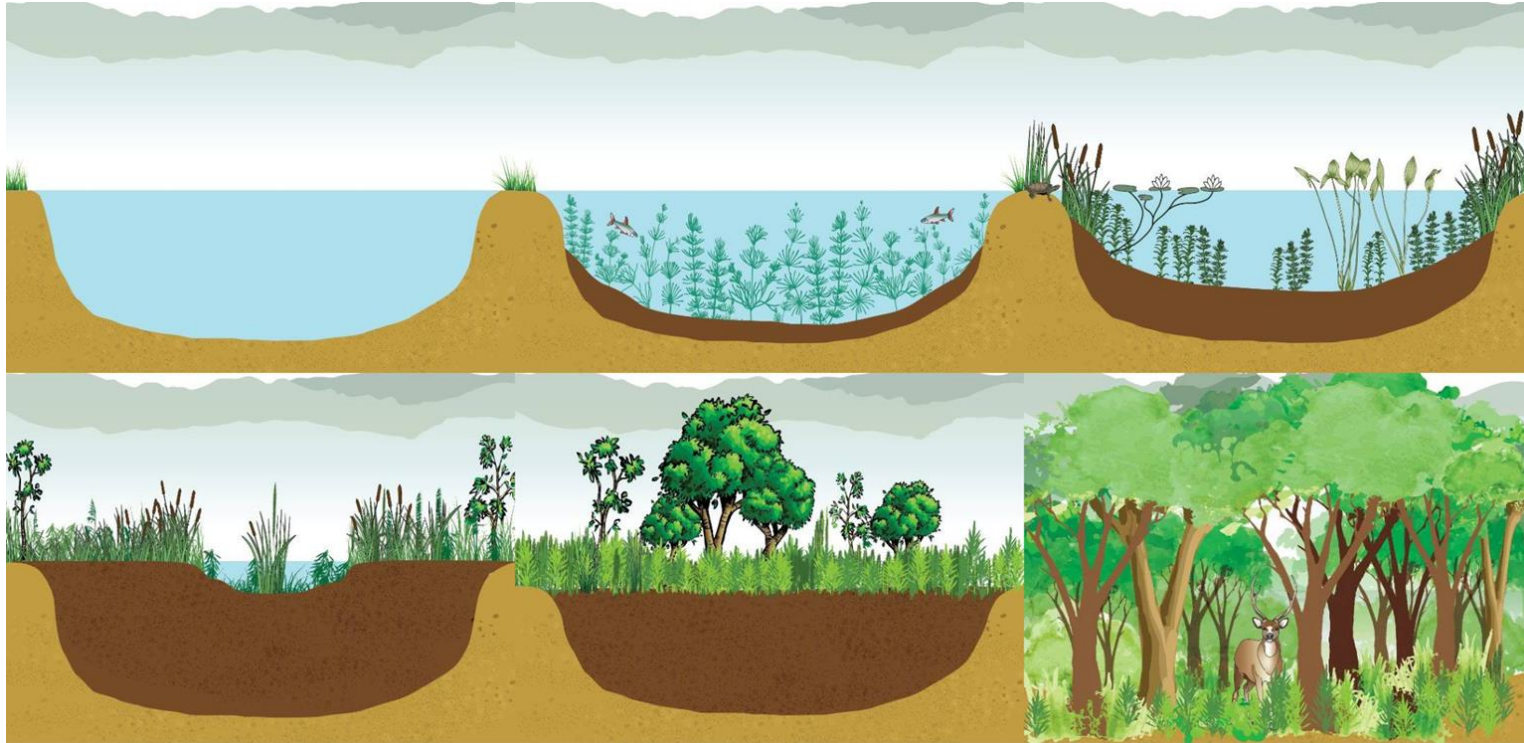
- Invading plants depend on condition of soil, availability of water, presence of other seeds or propagules.
- Primary succession takes much longer to proceed while secondary succession is much faster.
- Final result is the same : Mesic Community.

Ecological Succession : insert sketches

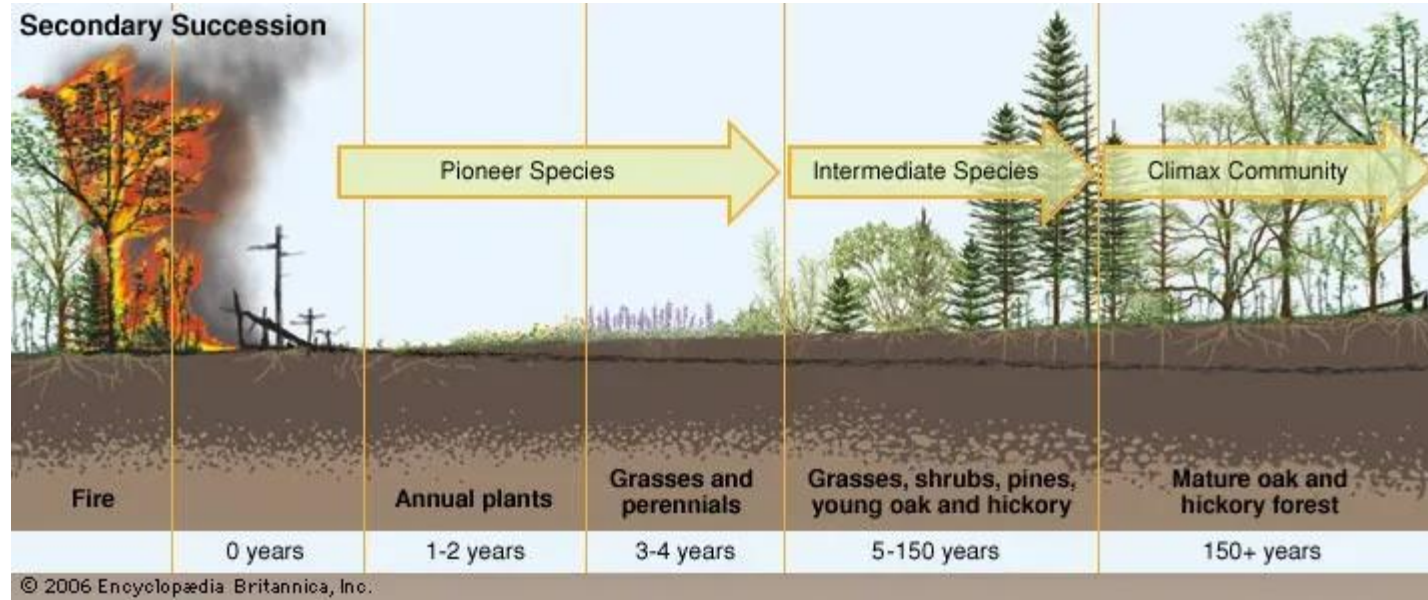
PRIMARY SUCCESSION



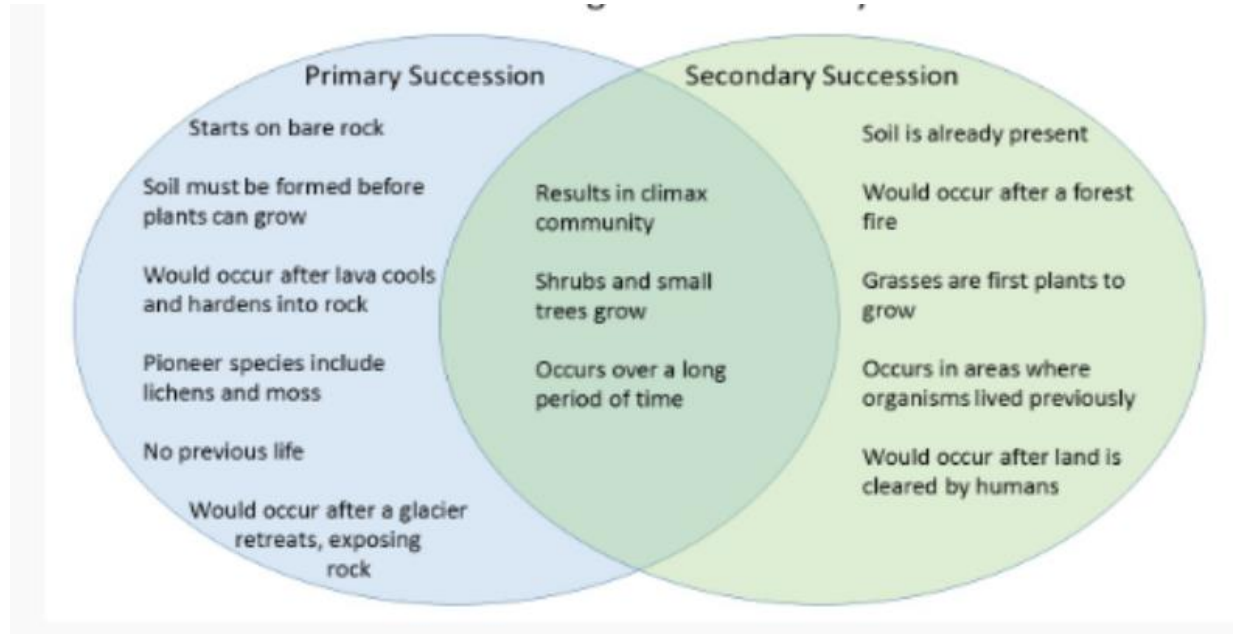
Ecological Succession



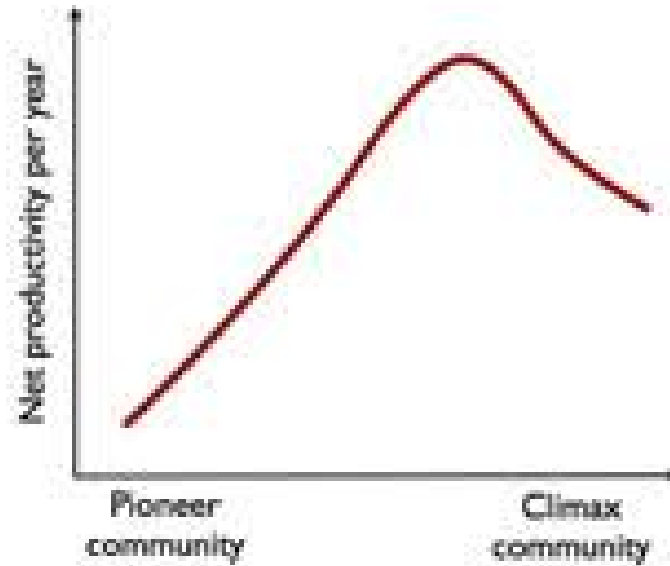
Ecological Succession : insert sketches



Ecological Succession



Ecological Succession



Ecological Succession



- Autogenic succession
 - Brought about by living inhabitants of that community itself
- Allogenic succession
 - Brought about by outside forces
- Autotrophic succession
 - Succession in which, initially the green plants dominate
- Heterotrophic
 - Succession in which, initially heterotrophs dominate

Ecological Succession



- Succession would be much faster in areas that exist in the middle of a large continent. All propagules or seeds belonging to different seers would reach here much faster and climax community would establish faster.

- Clement's theory of succession
 - 6 Phases:
 - Nudation : development of a bare site, disturbance
 - Migration : arrival of propagules
 - Ecesis : establishment of initial growth of vegetation.
 - Competition : as vegetation becomes well established, grows and spreads, various species compete for space, light and nutrients.

Ecological Succession



- Clement's theory of succession
 - Reaction : autogenic changes affect the habitat resulting in replacement of one plant community by another.
 - Stabilization : stable climax community

Biogeochemical Cycles



- Water Cycle
- Nitrogen Cycle
- Carbon Cycle
- Phosphorous Cycle
- Sulphur Cycle
- Oxygen Cycle

Biogeochemical Cycle



- Carbon, hydrogen, oxygen, nitrogen and phosphorus as elements and compounds make up 97% of the mass of our bodies and are more than 95% of the mass of all living organisms.
- These elements or mineral nutrients are always in circulation moving from non-living to living and then back to the non-living components of the ecosystem in a more or less circular fashion.
- This circular fashion is known as biogeochemical cycling
- Bio for living + Geo for earth.

Biogeochemical Cycle



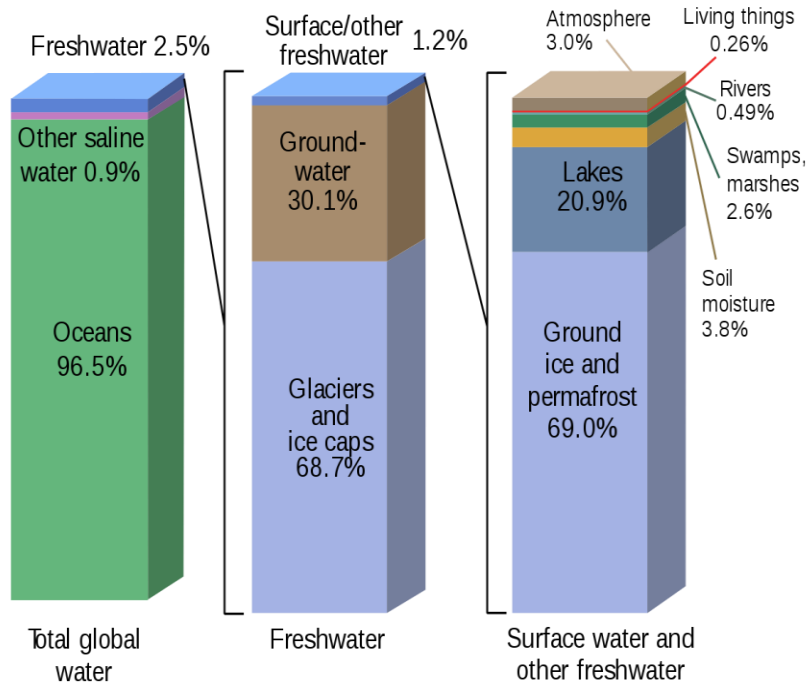
Gaseous Cycle	Sedimentary Cycle
Reservoir is atmosphere and ocean	Reservoir is earth's crust
Nutrient occurs as gas or vapour	Nutrients are non-gaseous
Comparatively quick and fast	Comparatively slow
Carbon, hydrogen, nitrogen cycles	Phosphorous and sulphur cycle

- Continuous movement of water on earth
- **Evaporation**- water evaporates from the surface of the earth and water bodies such as the oceans, seas, lakes etc
- **Sublimation**- Ice sheets and ice caps from north and south poles, and icecaps on mountains, get converted into water vapour directly, without converting into liquid.
- **Transpiration**- process by which plants release water vapour into the atmosphere through stomata in leaves and stems.
- **Condensation**- water vapour present at higher altitude condenses to form very tiny particles of water droplets. These particles come close together to form clouds and fog.

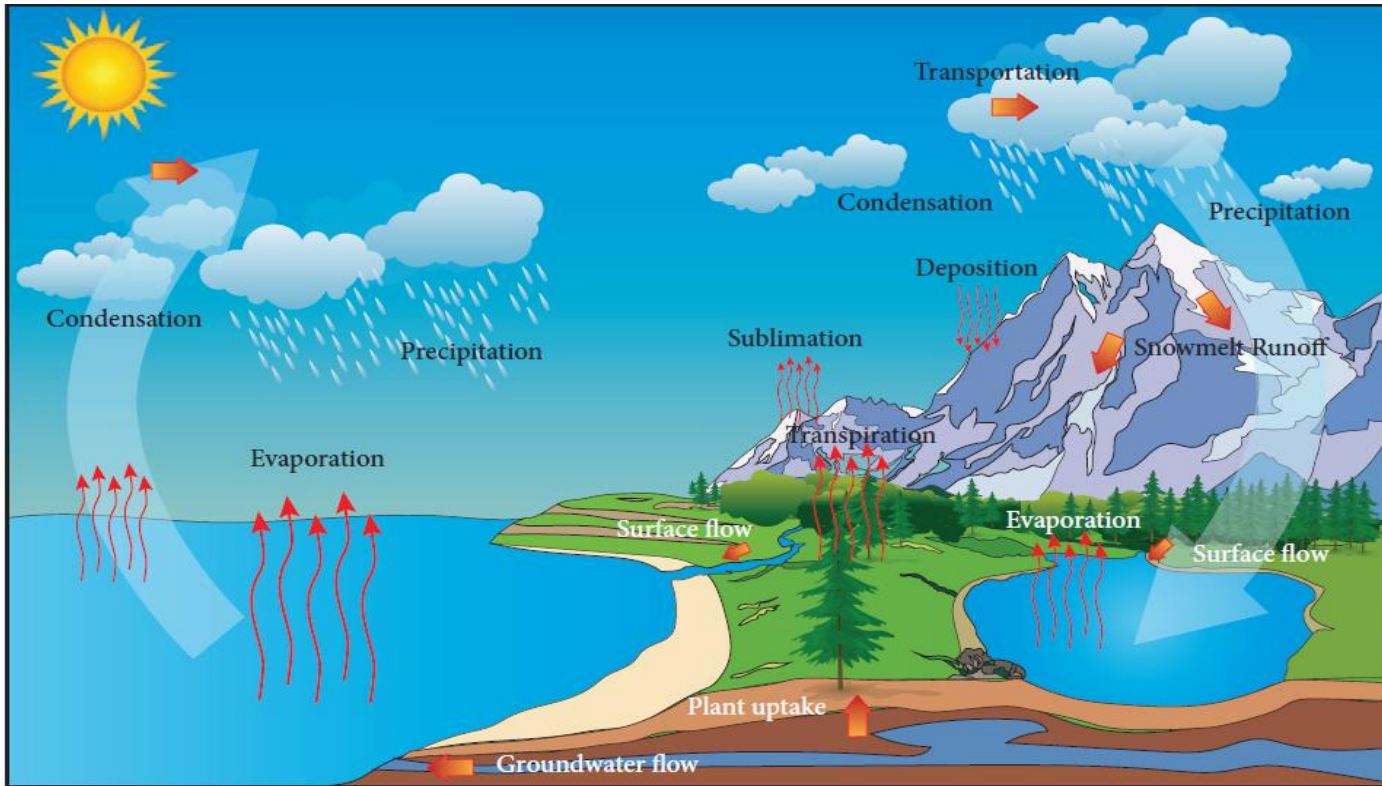
- **Precipitation**- clouds combine to make bigger droplets, and pour down as precipitation(rain).
- Precipitation includes drizzle, rain, snow and hail.
- **Surface runoff**- As the water pours down, it runs over the surface of earth.
- **Infiltration**- Some of the precipitated water moves deep into the soil. Then it moves down and increases the ground water level.
- **Percolation**- Some of the precipitated water flows through soil and porous or fractured rock

Water Cycle

Where is Earth's Water?



Water Cycle



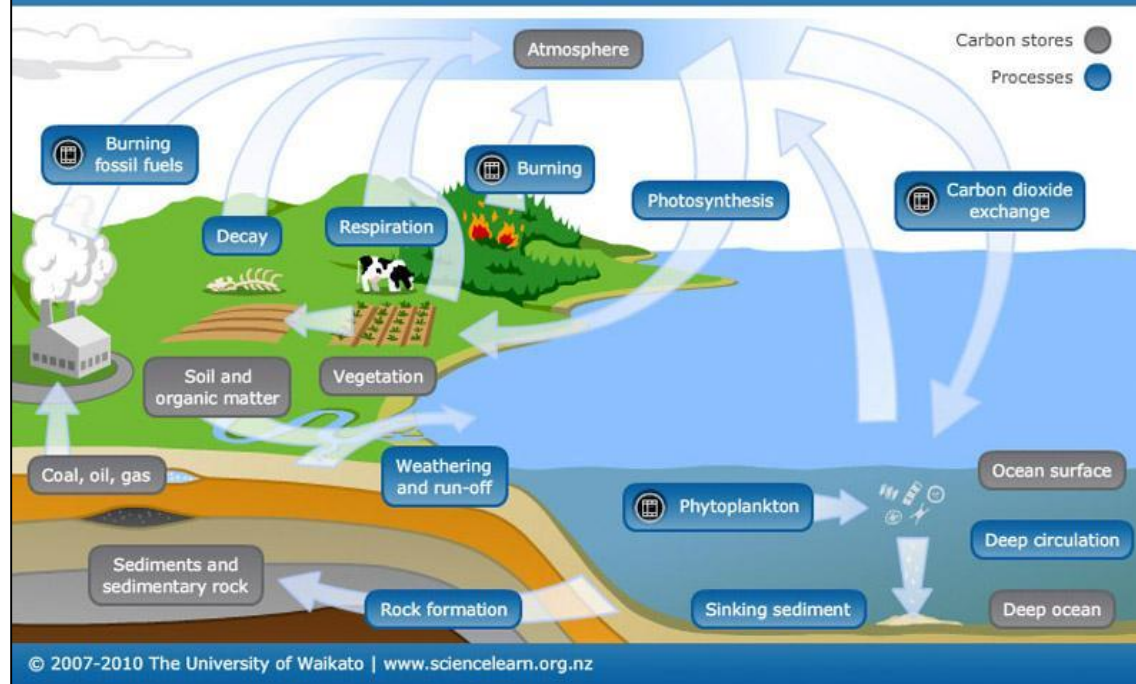
Carbon Cycle



- Carbon occurs in various forms on earth.
- Charcoal, diamond and graphite are elemental forms of carbon.
- Combined forms of carbon include carbon monoxide, carbon dioxide and carbonate salts.
- All living organisms are made up of carbon containing molecules like proteins and nucleic acids.
- Short term- photosynthesis, respiration
- Long term- fossil fuels formed by dead plants and animals

Carbon Cycle

CARBON CYCLE



Carbon Cycle



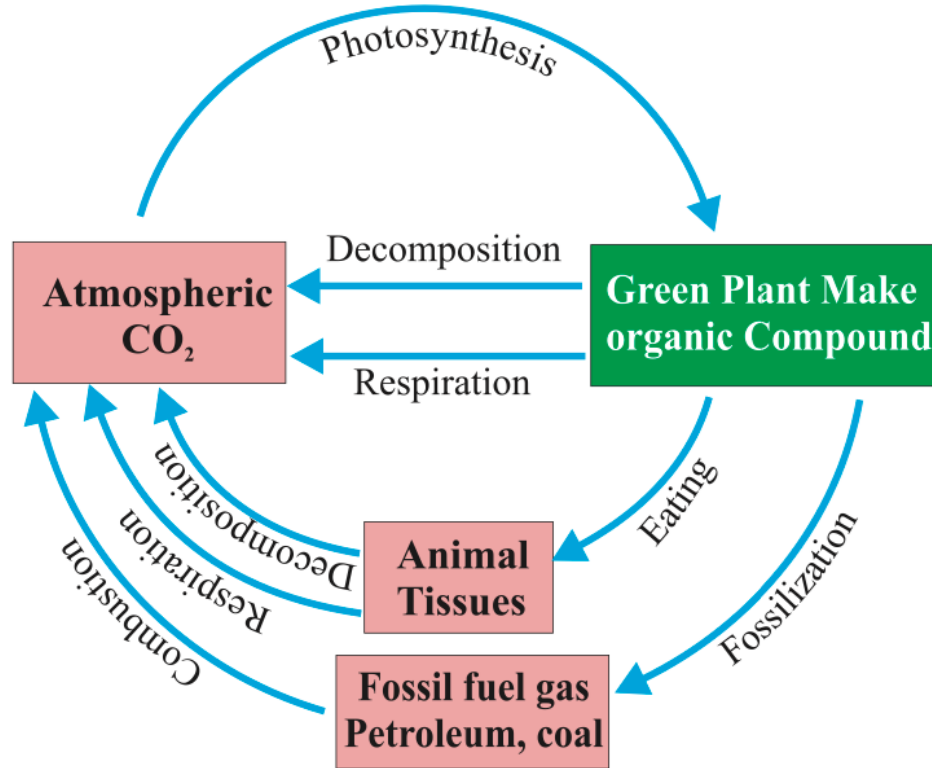
- Sources : Burning of fossil fuels release Carbon dioxide to the atmosphere.
- Natural Processes : Volcanic Eruption, Respiration, decay of organic matter.
- Sinks : Photosynthesis, Oceans (33% of atmospheric carbon dioxide is taken by the ocean, when too much carbon dioxide is emitted, it leads to ocean acidification), Dry ice formation at poles (climate change reduces this)

Carbon Cycle



- Short term : Photosynthesis, respiration, decomposition.
- Long term : accumulation as undecomposed organic matter in peaty layers of marshy soil or as insoluble carbonates in bottom sediments. Formation of fossil fuels such as coal, oil and natural gas. When they are burnt, carbon stored in them is released back to the air as carbon dioxide.

Carbon Cycle



Nitrogen Cycle



- Important nutrient needed for the survival of all living organisms. It is an essential component of proteins, DNA and chlorophyll.
- Atmosphere is a rich source of nitrogen and contains about 78% nitrogen.
- Plants and animals cannot utilize atmospheric nitrogen.
- They can use it only if it is in the form of ammonia, amino acids or nitrates. This conversion happens through nitrogen cycle.

- Nitrogen fixation-
 - Conversion of atmospheric nitrogen, which is in inert form, to reactive compounds available to living organisms.
 - Biological
 - Number of bacteria and blue green algae (Cyanobacteria) or nitrogen fixing bacteria Rhizobium which have symbiotic relationship with Leguminous plants like pea and beans
 - Lightning formation of NO_x and then Nitric Acid.
 - Chemical : Haber process- ammonia production

- Nitrogen fixation-
 - Biological :
 - Could be done by free living nitrifying bacteria : aerobic azotobacter and anaerobic clostridium
 - Or symbiotic nitrifying bacteria : living in association with leguminous plants and symbiotic bacteria living in non leguminous root nodules (rhizobium) or blue green algae (anabena, nostoc)

- Nitrification
 - Nitrosomonas bacteria promote transformation of ammonia into nitrite.
 - Nitrite is then further transformed into nitrate by the bacteria Nitrobacter.
- Nitrogen assimilation
 - Plants absorb nitrate ions and use them for making organic matter like amino acids, which are building blocks of proteins.
 - Herbivorous animals convert plant proteins into animal proteins. Carnivorous animals synthesize proteins from their food.

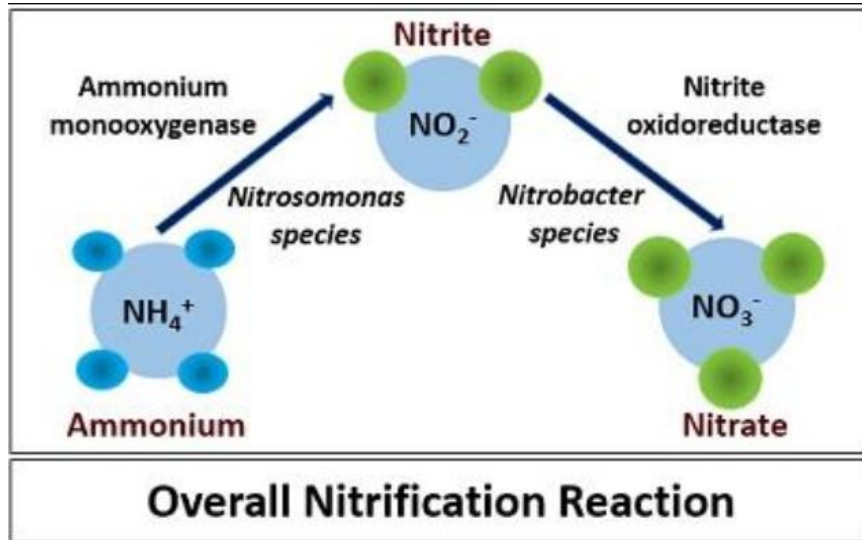
- Ammonification
 - The process of decomposition of nitrogenous waste by putrefying bacteria and fungi into ammonium compounds. From excretion / death of animals, nitrogen is returned to the soil.
- Denitrification
 - Free living soil bacteria such as *Pseudomonas* sp. reduce nitrate ions of soil into gaseous nitrogen which enters the atmosphere.

Nitrogen Cycle



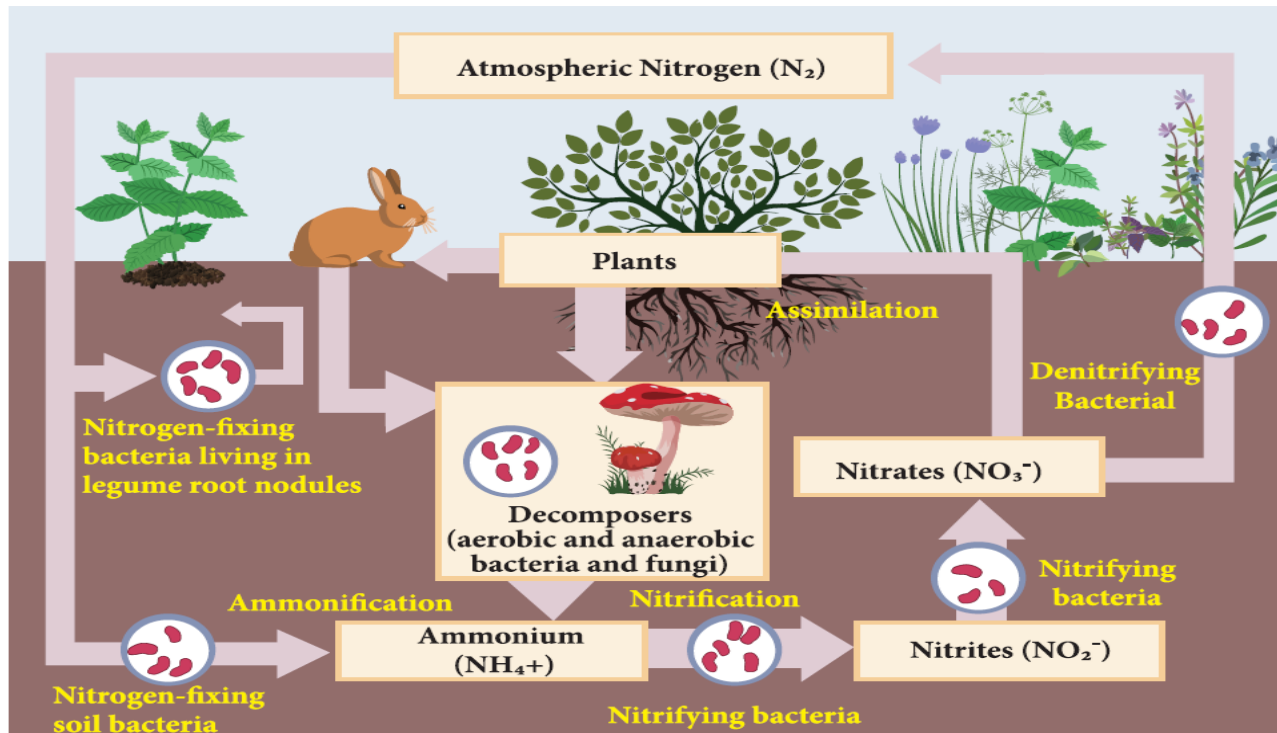
- Certain quantity of soil nitrates being highly soluble in water, is lost to the system by being transported away by surface run off.

Nitrogen Cycle



Role played in nitrogen cycle	Name of the microorganisms
Nitrogen fixation	<i>Azotobacter</i> (in soil) <i>Rhizobium</i> (in root nodules) Blue green algae- <i>Nostoc</i>
Ammonification	Putrefying bacteria, Fungi
Nitrification	Nitrifying bacteria i. <i>Nitrosomonas</i> ii. <i>Nitrobacter</i>
Denitrification	Denitrifying bacteria <i>Pseudomonas</i>

Nitrogen Cycle



Phosphorous Cycle



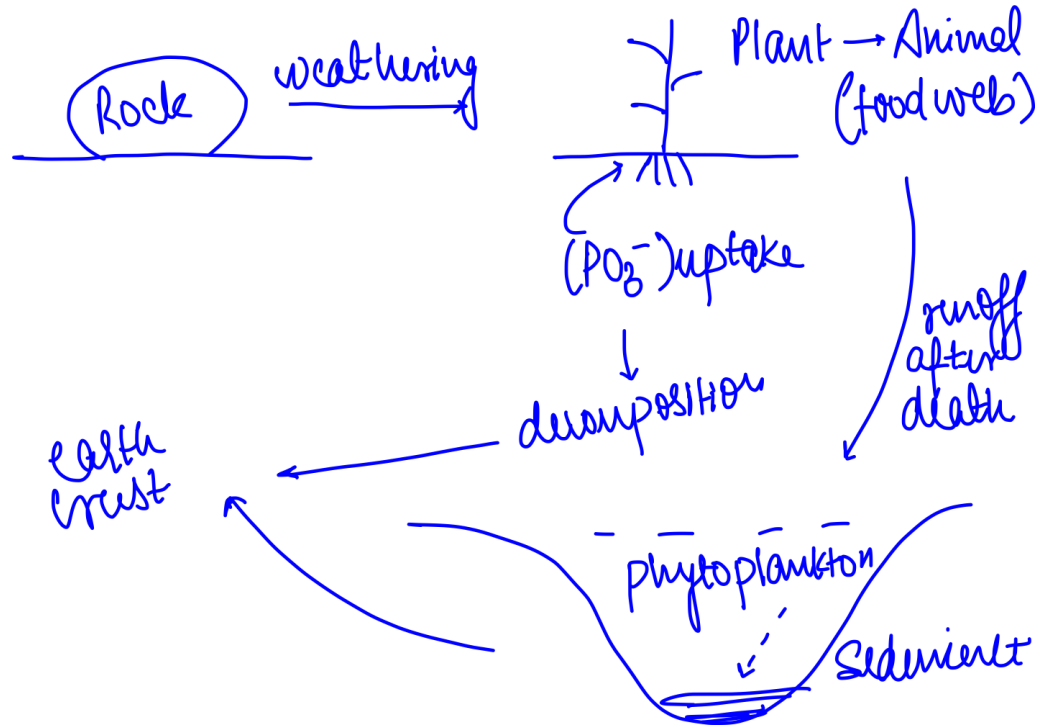
- Phosphorus plays a central role in aquatic ecosystems and water quality.
- Unlike carbon and nitrogen, which come primarily from the atmosphere, phosphorus occurs in large amounts as a mineral in phosphate rocks and enters the cycle from erosion and mining activities.
- This is the nutrient considered to be the main cause of excessive growth of rooted and free-floating microscopic plants in lakes.
- The main storage for phosphorus is in the earth's crust.

Phosphorous Cycle



- On land phosphorus is usually found in the form of phosphates.
- By the process of weathering and erosion phosphates enter rivers and streams that transport them to the ocean.
- In the ocean once the phosphorus accumulates on continental shelves in the form of insoluble deposits.
- After millions of years, the crustal plates rise from the sea floor and expose the phosphates on land.
- After more time, weathering will release them from rock and the cycle's geochemical phase begins again.

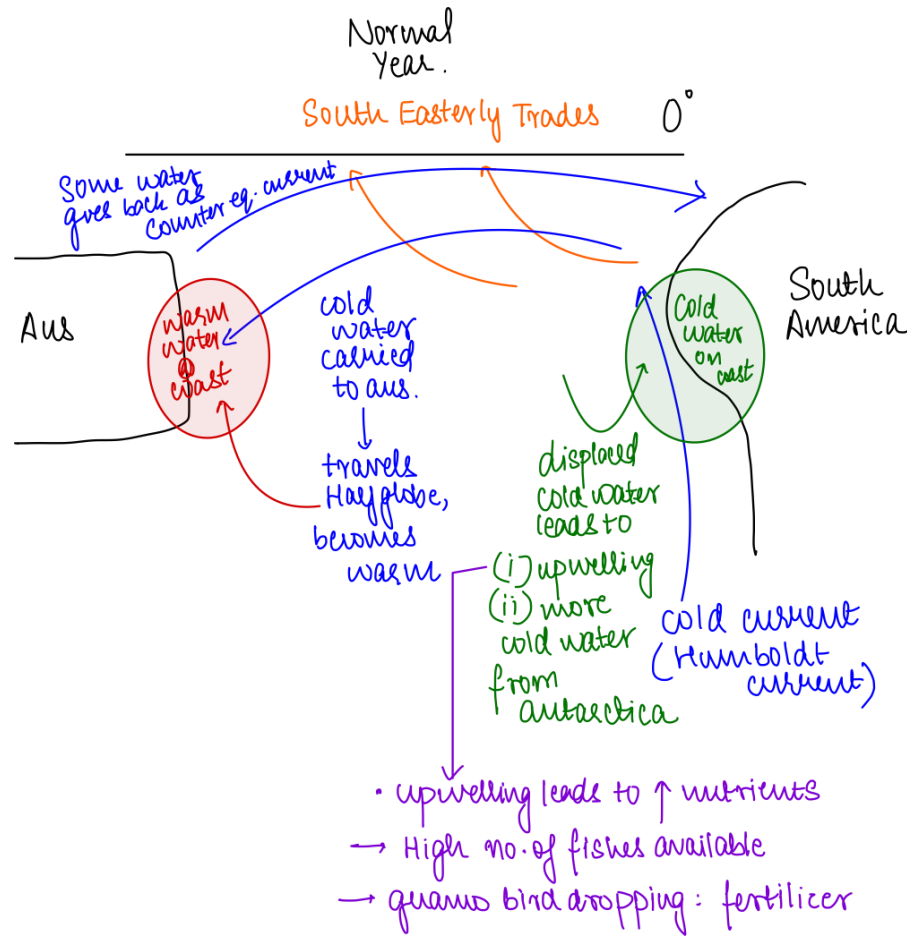
Phosphorous Cycle



Phosphorous Cycle

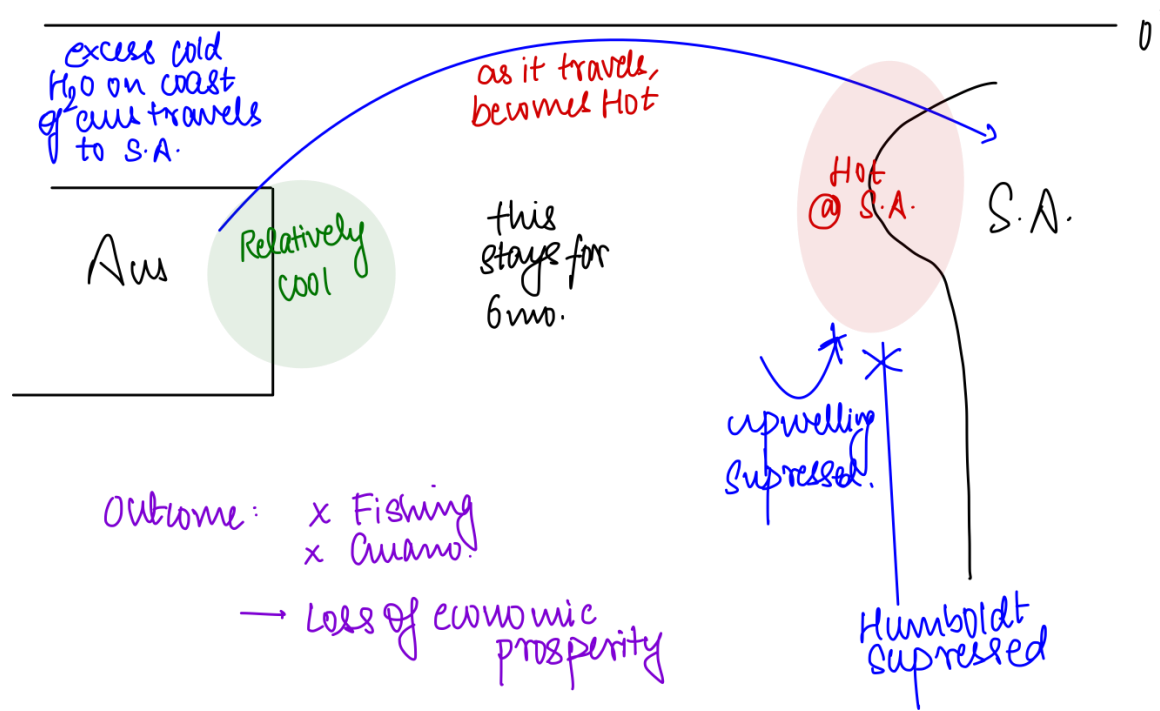


- Guano bird droppings on the coast of ecuador contain phosphorous, act as feritizer.
- El nino and La nina : relate with geography.
- Upwelling at coast of ecuador and peru leads to increased presence of fish, leading to more guano bird droppings.

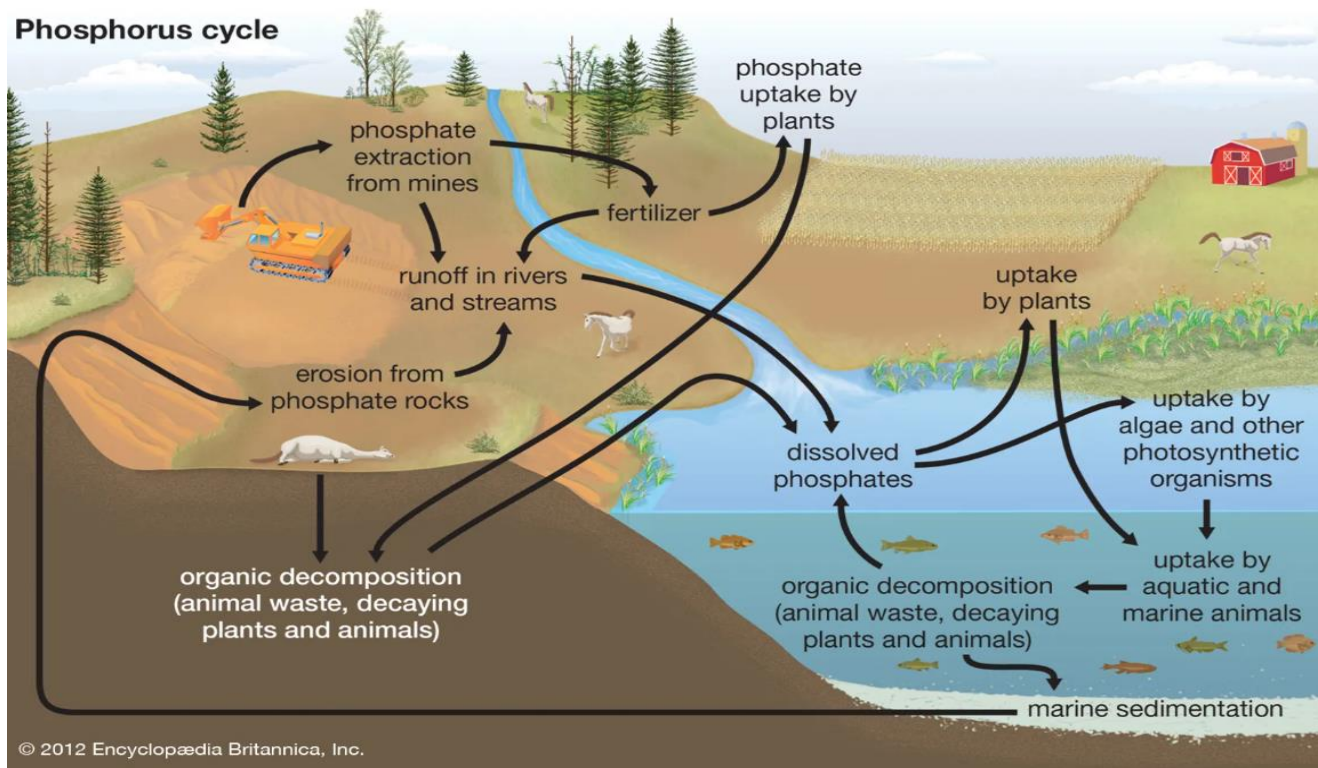


EL-nino

arrival of first child
(close to Christmas)



Phosphorous Cycle



- Reservoir
 - Soil and sediments in Organic (coal, oil and peat) and Inorganic deposits (pyrite rock and sulphur rock)
 - Form- sulphates, sulphides and organic sulphur.
- Released by weathering of rocks, erosional runoff and decomposition of organic matter and is carried to terrestrial and aquatic ecosystems in salt solution.

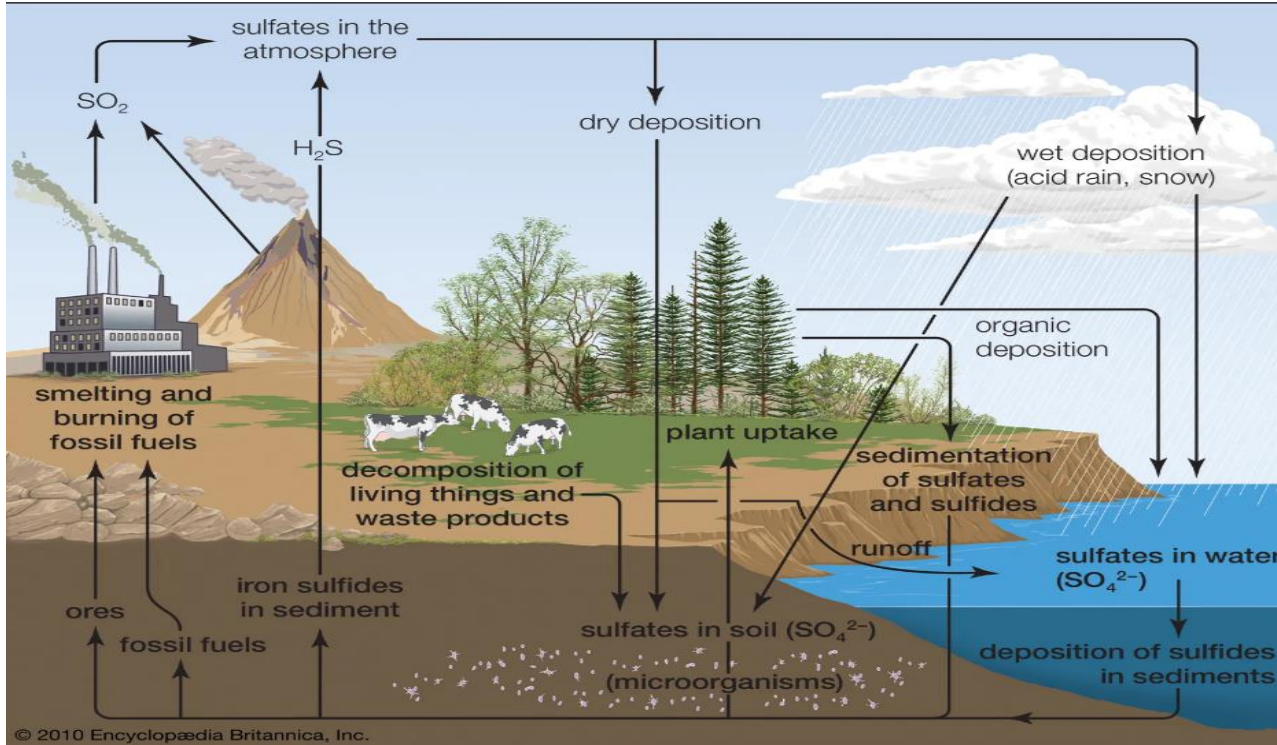
- Largely sedimentary, however enter atmosphere through-
 - Volcanic eruptions, combustion of fossil fuels, from surface of ocean and from gases released by decomposition
 - Forms- Hydrogen sulphide (H_2S) and sulphur dioxide (SO_2)
 - Atmospheric hydrogen sulphide also gets oxidised into sulphur dioxide.
 - Atmospheric sulphur dioxide is carried back to the earth after being dissolved in rainwater as weak sulphuric acid

Sulphur Cycle



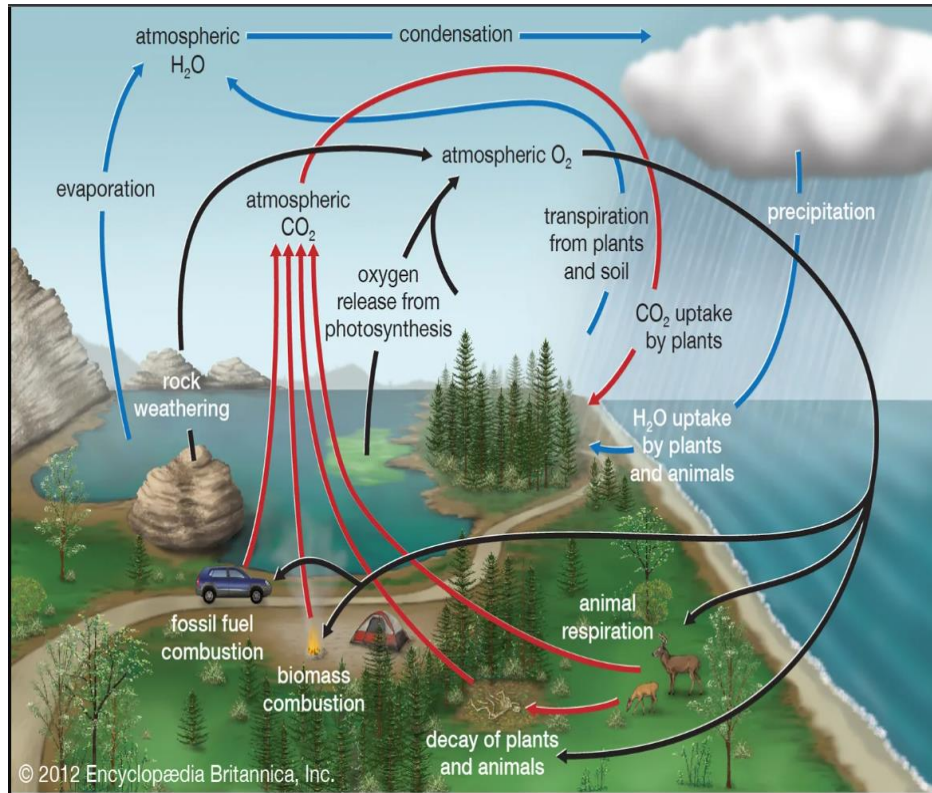
- Sulphur in the form of sulphates is taken up by plants and incorporated through a series of metabolic processes into sulphur bearing amino acids.
- From living organisms, it is carried back to the soil and to the bottom of ponds and lakes through excretion and decomposition of dead organic materials.

Sulphur Cycle



- 21% in air in atmosphere= Oxygen
- Oxygen constitutes about 65% of the mass of the human body and most of this is in the form of water.
- The steps involved in the oxygen cycle are:
- Stage-1: All green plants during the process of photosynthesis, release oxygen back into the atmosphere as a by-product.
- Stage-2: All aerobic organisms use free oxygen for respiration.
- Stage-3: Animals exhale Carbon dioxide back into the atmosphere which is again used by the plants during photosynthesis. Now oxygen is balanced within the atmosphere.

Oxygen Cycle



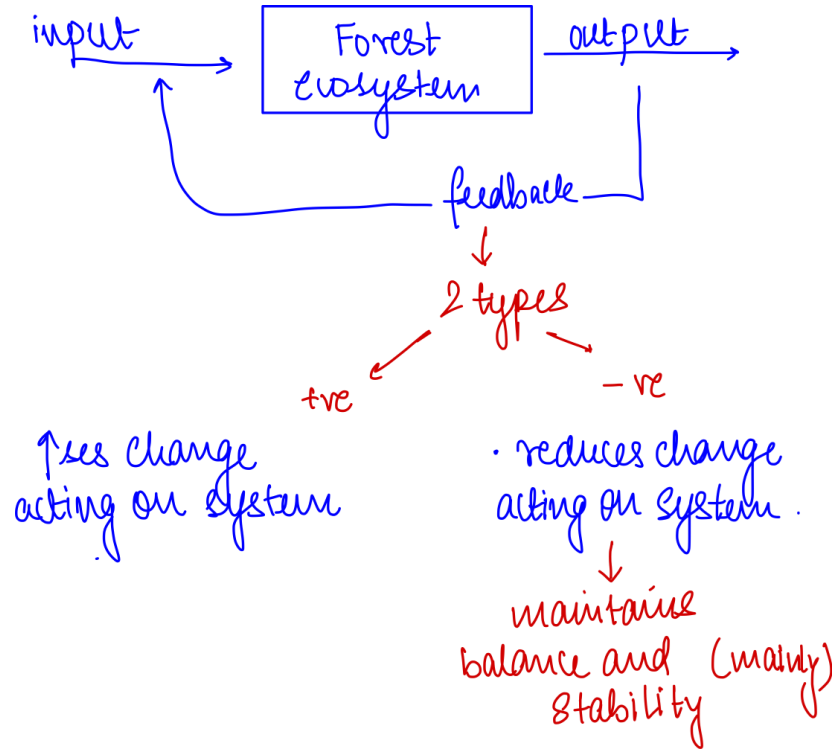
Homeostasis of Ecosystem



- An ecosystem maintains a biological equilibrium between the different components
- It keeps on changing with the time and is not static.
- Feedback system
 - One component of ecosystem keeps a check on the population of the other component

Homeostasis of Ecosystem

- An ecosystem is a self regulating system that maintains equilibrium and steady state through feedbacks.



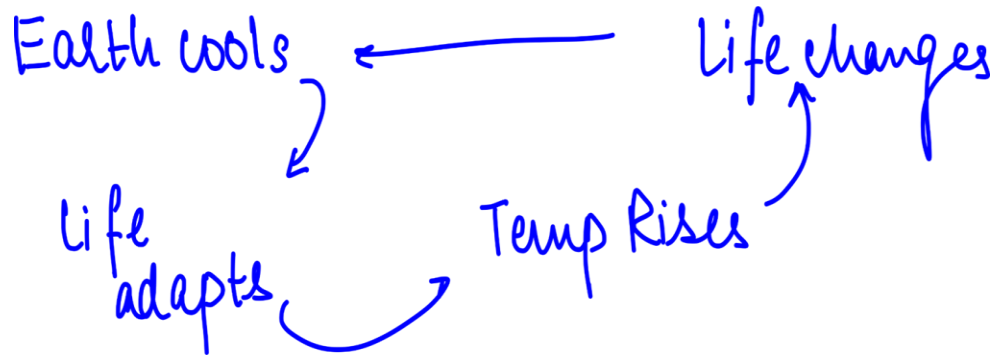
- Positive feedback
 - The increase in the population of the organisms at the different levels increases the population of organisms at a lower level
 - For example, when of plants increases it leads to increase in the population of herbivore animals. It increases the population of frogs and birds.
- Negative feedback
 - The increased population of insectivorous animals acts on the herbivorous insect by the process of predation.

Gaia Hypothesis



- It was given by by British chemist James E. Lovelock and U.S. biologist Lynn Margulis.
- It proposes that living organisms interact with their inorganic surroundings on Earth to form a synergistic and self-regulating, complex system that helps to maintain and perpetuate the conditions for life on the planet.

Gaia Hypothesis



What is Gaia?



Greek mythology – goddess of the earth
mother of living things, sustainer of life

Biosphere, Atmosphere and Hydrosphere along with evolution of life maintains homeostasis on earth.

Global Temperature - Ocean Salinity - Oxygen in Atmosphere - Habitability

Existence of a cybernetic feedback mechanism unconsciously operated by the biota.

For example : ocean salinity is fixed at 3.4% due to bacterial processes even though there is constant deposition of salt by rivers.

Daisy world : Mathematical Model to predict this.

- Healthy ecosystems are the base for a wide range of economic, environmental and aesthetic goods and services. The products of ecosystem processes are named as ecosystem services.
- Researchers have put an average price tag of US \$ 33 trillion a year on these fundamental ecosystems services

Ecosystem Services



Provisioning Services

are the material benefits people get from ecosystems for e.g. supply of food, water, fibers, wood and fuels.



Regulating Services

are the benefits obtained from the regulation of ecosystem processes e.g. the regulation of air quality and soil fertility, control of floods or crop pollination.



Supporting Services

are necessary for the production of all other ecosystem services, for e.g. by providing plants and animals with living spaces, allowing for diversity of species, and maintaining genetic diversity.



Cultural Services

are non-material benefits people gain from ecosystems, for e.g. aesthetic and engineering inspiration, cultural identity and spiritual well-being.

Ecosystem Services



Ecosystem related Initiatives



India : Cultural services : the idea of sacred groves.

Using ecosystem services, we calculate the true value of ecosystems. For example, when we look at an afforestation program, we look at not only the benefit of timber, but also other benefits.

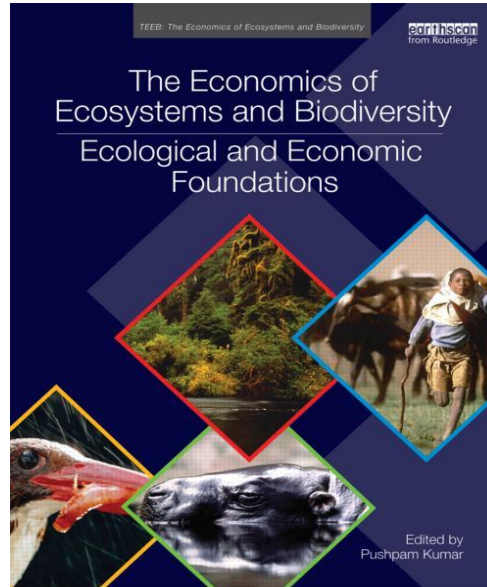
These are the consequences of capitalism intermixing with ecology, where we are assigning monetary value to ecology. This is an example of shallow ecology.

Deep ecology : preservation for preservation's sake.

- The Economics of Ecosystems and Biodiversity (TEEB)
 - Global initiative focused on “making nature’s values visible”.
 - Its principal objective is to mainstream the values of biodiversity and ecosystem services into decision-making at all levels.
 - Captures monetary valuation of ecosystem services and biodiversity for Green Domestic Product
 - Concept was given by Pavan Sukhdev, received Tyler Prize (Nobel equivalent for environment) in 2020.

- Global initiative of UNEP, G8, BASIC, Mexico
- Looks into green accounting methodology to calculate green domestic product.
- Creates responsibility in stakeholders for conservation of ecosystems and biodiversity.
- Part of UNEP green economy initiative.
- Economic liability is assigned to forests on the basis of services provided.

Ecosystem related Initiatives



Payment of Ecosystem Services



- Payment for Ecosystem Services (PES) is a concept that aims to provide financial incentives to individuals or communities for the preservation, restoration, or enhancement of ecosystems and the services they provide.
- It is a market-based approach that recognizes the value of ecosystem services, such as clean water, carbon sequestration, and biodiversity conservation.
- Mechanism: Direct payments to landowners or communities, or through payments for environmental services such as carbon credits or water quality credits.

Payment of Ecosystem Services



- Ecological footprint
 - Ecological Footprint accounting measures a population's demand for natural ecosystems' supply of resources and services.
 - On the demand side- measures an individual or a population's demand for plant-based food and fiber products, livestock and fish products, timber and other forest products, space for urban infrastructure, and forest to absorb its carbon dioxide emissions from fossil fuels.
 - On the supply side, a city, state, or nation's biocapacity represents its biologically productive land and sea area, including forest lands, grazing lands, cropland, fishing grounds, and built-up land.

- What is Earth Overshoot Day?
 - When the entire planet is running an ecological deficit, we call it “overshoot.”
At the global level, ecological deficit and overshoot are the same, since there is no net import of resources to the planet.
 - Earth Overshoot Day marks the date when humanity’s demand for ecological resources and services (Ecological Footprint) in a given year exceeds what Earth can regenerate in that year (biocapacity).
 - 2022- 28th July vs 1970- 30th December

Test of 12 Questions!

Cut off : 10/12



Q.1 (2019)

Q.) Consider the following statements:

1. Some species of turtles are herbivores.
2. Some species of fish are herbivores.
3. Some species of marine mammals are herbivores.
4. Some species of snakes are viviparous.

Which of the statements given above are correct?

- a) 1 and 3 only
- b) 2, 3 and 4 only
- c) 2 and 4 only
- d) 1, 2, 3 and 4

Q.2 (2021)



Q.) In the case of which of the following biogeochemical cycles, the weathering of rocks is the main source of release of nutrients to enter the cycle?

- (a) Carbon cycle
- (b) Nitrogen cycle
- (c) Phosphorus cycle
- (d) Sulphur cycle

1. The amount of water in the rivers and lakes is more than the amount of groundwater.

2. The amount of water in polar ice caps and glaciers is more than the amount of groundwater.

Which of the statements given above is/are correct?

- (a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Q.4 (2017)

Q.) With reference to an initiative called 'The Economics of Ecosystems and Biodiversity (TEEB)', which of the following statements is/are correct?

1. It is an initiative hosted by UNEP, IMF and World Economic Forum
2. It is a global initiative that focuses on drawing attention to the economic benefits of biodiversity
3. It presents an approach that can help decision-makers recognize, demonstrate and capture the value of ecosystems and biodiversity

Select the correct answer using the code given below.

- | | |
|------------------|----------------|
| (a) 1 and 2 only | (b) 3 only |
| (c) 2 and 3 only | (d) 1, 2 and 3 |

Q.5 (2014)



Q.) Which one of the following is the process involved in photosynthesis?

- (a) Potential energy is released to form free energy
- (b) Free energy is converted into potential energy and stored
- (c) Food is oxidized to release carbon dioxide and water.
- (d) Oxygen is taken, and carbon dioxide and water vapour are given out

Q.6 (2013)

Q.) On the planet earth, most of the freshwater exists as ice caps and glaciers. Out of the remaining freshwater, the largest proportion

- (a) is found in atmosphere as moisture and clouds
- (b) is found in freshwater lakes and rivers
- (c) exists as groundwater
- (d) exists as soil moisture

Q.7 (2013)

Q.) Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity?

- (a) Oceans, lakes, grasslands, mangroves
- (b) Mangroves, oceans, grasslands, lakes
- (c) Mangroves, grasslands, lakes, oceans
- (d) Oceans, mangroves, lakes, grasslands

Q. 8 (2013)

Q.) Which of the following adds / add nitrogen to the soil?

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below.

- A. 1 only
- B. 2 and 3 only
- C. 1 and 3 only
- D. 1, 2 and 3

Q. 9 (2012)

The Millennium Ecosystem Assessment describes the following major categories of ecosystem services provisioning, supporting, regulating, preserving and cultural.

Which one of the following is supporting service?

- (a) Production of food and water
- (b) Control of climate and disease
- (c) Nutrient cycling and crop pollination
- (d) Maintenance of diversity

Q. 10 (2014)

Which of the following adds/add carbon dioxide to the carbon cycle on the planet Earth?

1. Volcanic action
2. Respiration
3. Photosynthesis
4. Decay of organic matter

Select the correct answer using the code given below.

- a) 1 and 3 only
- b) 2 only
- c) 1, 2 and 4 only
- d) 1, 2, 3 and 4

Q. 11 (2014)

Consider the following statements:

1. Agricultural soils release nitrogen oxides into environment.
2. Cattle release ammonia into environment.
3. Poultry industry releases reactive nitrogen compounds environment.

Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 and 3 only
- (c) 2 only
- (d) 1, 2 and 3

Q. 12 (2021)

Q4. In the nature, which of the following is/are most likely to be found surviving on a surface without soil?

1. Fern
2. Lichen
3. Moss
4. Mushroom

Select the correct answer using the code given below.

- (a) 1 and 4 only
- (b) 2 only
- (c) 2 and 3
- (d) 1, 3 and 4

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1 (2019)

Q.) Consider the following statements:

1. Some species of turtles are herbivores.
2. Some species of fish are herbivores.
3. Some species of marine mammals are herbivores.
4. Some species of snakes are viviparous.

Which of the statements given above are correct?

- a) 1 and 3 only
- b) 2, 3 and 4 only
- c) 2 and 4 only
- d) **1, 2, 3 and 4**

Answer: D

- Adult sea turtles are herbivores — unlike most sea turtle species — and their serrated jaws help them chew vegetation such as algae and seagrass.
- Herbivorous fishes are fishes that eat plant material. Surgeonfish and parrotfish are two familiar examples, often seen browsing and scraping on reef algae.



- Manatees, often called sea cows, are voracious herbivores who eat up to one-tenth of their massive body weight every 24 hours. Sea cows are mammals.
- Snakes that are viviparous nourish their developing young through a placenta and yolk sac, something that is highly unusual among reptiles. Boa constrictors and green anacondas are two examples of viviparous snakes, meaning they give birth to live young with no eggs involved at any stage of development.



Q.2 (2021)

Q.) In the case of which of the following biogeochemical cycles, the weathering of rocks is the main source of release of nutrients to enter the cycle?

- (a) Carbon cycle
- (b) Nitrogen cycle
- (c) Phosphorus cycle**
- (d) Sulphur cycle

Answer: C

- Carbon and Nitrogen cycle are the gaseous cycles. Here, the main reservoir of nutrient is the atmosphere or the hydrosphere.
- In sulphur cycle, nutrient is released by weathering of rocks, erosional runoff and decomposition of organic matter. The sulphur cycle is mostly sedimentary but two of its compounds hydrogen sulphide (H_2S) and sulphur dioxide (SO_2) add a gaseous component to its normal sedimentary cycle.
- In Phosphorus cycle, nutrients got released mainly by weathering of rocks since the Phosphorus mainly occurs as a mineral in phosphate rocks. It enters the cycle from erosion and mining activities.

Answer: **B**

- Statement 1 is incorrect. Of the total freshwater, 69% resides in glaciers, 30% underground, and less than 1% is located in lakes, rivers, and swamps. Thus, the amount of water in rivers and lakes is less than amount of groundwater.
- Statement 2 is correct. Of the total freshwater, 69% resides in glaciers and 30% underground. Polar ice caps and glaciers has more water than groundwater

Q.4 (2017)

Q.) With reference to an initiative called 'The Economics of Ecosystems and Biodiversity (TEEB)', which of the following statements is/are correct?

1. It is an initiative hosted by UNEP, IMF and World Economic Forum
2. It is a global initiative that focuses on drawing attention to the economic benefits of biodiversity
3. It presents an approach that can help decision-makers recognize, demonstrate and capture the value of ecosystems and biodiversity

Select the correct answer using the code given below.

(a) 1 and 2 only

(b) 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

Answer: C

- TEEB is a global initiative focused on “making nature’s values visible”. Its principal objective is to mainstream the values of biodiversity and ecosystem services into decision-making at all levels. TEEB demonstrates their values in economic terms and not its intrinsic value.

Q.5 (2014)

Q.) Which one of the following is the process involved in photosynthesis?

(a) Potential energy is released to form free energy

(b) Free energy is converted into potential energy and stored

(c) Food is oxidized to release carbon dioxide and water.

(d) Oxygen is taken, and carbon dioxide and water vapour are given out

Answer: **B**

- During photosynthesis, solar energy converted to chemical energy and stored in sugar molecule bonds. This chemical bond energy is potential (stored) energy.
- The correct answer is Food is oxidized to release carbon dioxide and water.
- Photosynthesis- It is the process by which green plants and certain other organisms transform light energy into chemical energy. Photosynthesis uses light energy to convert carbon dioxide and water into glucose and oxygen gas.

Q.6 (2013)

Q.) On the planet earth, most of the freshwater exists as ice caps and glaciers. Out of the remaining freshwater, the largest proportion

(a) is found in atmosphere as moisture and clouds

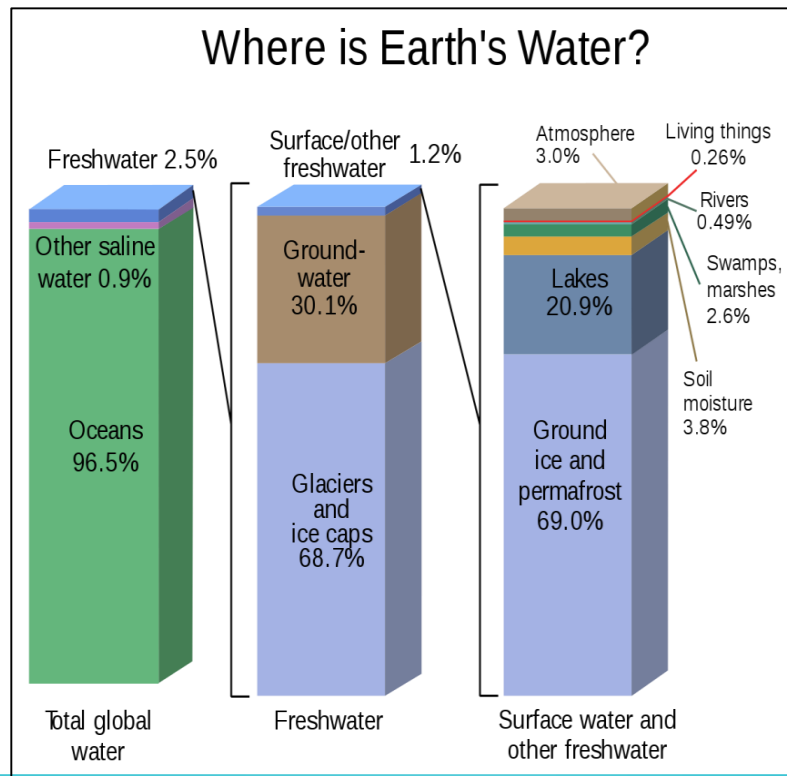
(b) is found in freshwater lakes and rivers

(c) exists as groundwater

(d) exists as soil moisture

Answer: C

- exists as groundwater
- Refer to earlier slides



Q.7 (2013)

Q.) Which one of the following is the correct sequence of ecosystems in the order of decreasing productivity?

- (a) Oceans, lakes, grasslands, mangroves
- (b) Mangroves, oceans, grasslands, lakes
- (c) Mangroves, grasslands, lakes, oceans**
- (d) Oceans, mangroves, lakes, grasslands

Answer: C

- Estuaries, swamps, marshes are all found in abundant in Mangrove forest, also dry matter produced per unit of area per land is highest in mangroves than other forest.
- Lakes grow because of the growth of plankton every day & also detritus. Initial growth in a grass land is high, but it tapers off.
- The average ocean productivity is about 50 grams of carbon per square meter per year.
- Average land productivity is 160 grams of carbon per square meter per year.
- The productivity of the open ocean (ocean away from coasts) is comparable to desert production.

Q.8 (2013)

Q.) Which of the following adds / add nitrogen to the soil?

1. Excretion of urea by animals
2. Burning of coal by man
3. Death of vegetation

Select the correct answer using the codes given below.

- A. 1 only
- B. 2 and 3 only
- C. 1 and 3 only**
- D. 1, 2 and 3

Answer: C

- Nitrogenous Waste products of Living organisms such as urea and uric acid and dead remains of organisms are converted back to inorganic ammonia by the bacteria
- Animal Waste like Urea, Uric acid and Death of vegetation add nitrogen in the form of nitrates directly into the soil.

Q. 9 (2012)

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Which one of the following is supporting service?

- (a) Production of food and water
- (b) Control of climate and disease
- (c) Nutrient cycling and crop pollination**
- (d) Maintenance of diversity

Answer: C

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- d) 1, 2, 3 and 4

Answer: C

Q. 11 (2014)

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Which of the statements given above is/are correct?

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- (c) 2 only
- (d) 1, 2 and 3**

Answer: D

Q. 12 (2021)

Q. In the nature, which of the following is/are most likely to be found surviving on a surface without soil?

1. Fern
2. Lichen
3. Moss
4. Mushroom

Select the correct answer using the code given below.

- (a) 1 and 4 only
(b) 2 only
(c) 2 and 3
(d) 1, 3 and 4

Answer: C